Program No : 6

Title: Write a program in C++ to perform following operations on complex numbers Add,Subtract, Multiply, Divide. Use operator overloading for these operations. The objective of this assignment is to learn the concepts of operator overloading.

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Program:

```
#include <iostream>

using namespace std;

class Complex

{

private:

   float real, img;

public:

   void accept(float r, float i)

   {

      real = r;

      img = i;

   }

   void display()

   {

      if (img >= 0)
```

```
        cout << real << " + " << img << "i" << endl;
    else
        cout << real << " - " << -img << "i" << endl;
}
Complex operator+(Complex b)
{
    Complex result;
    result.real = real + b.real;
    result.img = img + b.img;
    return result;
}
Complex operator-(Complex b)
{
    Complex result;
    result.real = real - b.real;
    result.img = img - b.img;
    return result;
}
Complex operator*(Complex b)
{
```

```
        Complex result;
        result.real = (real * b.real) - (img * b.img);
        result.img = (real * b.img) + (img * b.real);
        return result;
    }
    Complex operator/(Complex b)
    {
        Complex result;
        float denominator = (b.real * b.real) + (b.img * b.img);
        result.real = ((real * b.real) + (img * b.img)) / denominator;
        result.img = ((img * b.real) - (real * b.img)) / denominator;
        return result;
    }
};
int main()
{
    Complex C1, C2, C3;

    C1.accept(5, 3);
    C2.accept(5, 8);
```

```
cout << "First complex number: ";

C1.display();

cout << "Second complex number: ";

C2.display();



C3 = C1 + C2;

cout << "\nAddition: ";

C3.display();



C3 = C1 - C2;

cout << "Subtraction: ";

C3.display();



C3 = C1 * C2;

cout << "Multiplication: ";

C3.display();



C3 = C1 / C2;

cout << "Division: ";

C3.display();
```

```
    return 0;

}
```

Output:

Output

```
First complex number: 6 + 4i
Second complex number: 3 + 8i

Addition: 9 + 12i
Subtraction: 3 - 4i
Multiplication: -14 + 60i
Division: 0.684932 - 0.493151i
```